

Summary of the CTE DB-HE 2019

Application Guide

This document summarizes the application guide for the Basic Document on Energy Saving (DB-HE) of the Technical Building Code (CTE). It establishes the requirements for buildings to achieve reduced energy consumption, largely covered by renewable sources .

HE0: Limitation of Energy Consumption

This section limits the Total Primary Energy Consumption ($C_{ep,tot}$) and Non-Renewable Primary Energy Consumption ($C_{ep,nren}$) to mitigate climate change and reduce energy dependency .

1. General Calculation Concepts

The final energy consumption depends on the demand and the efficiency of the systems:

$$C_{final} = \frac{\text{Energy Demand}}{\text{Mean System Efficiency } (\eta)}$$

2. Internal Heat Load (CFI)

For tertiary use buildings, the limit of consumption depends on the internal load, calculated for a standard week:

$$CFI = \frac{\sum C_{oc}}{7 \cdot 24} + \frac{\sum C_{il}}{7 \cdot 24} + \frac{\sum C_{eq}}{7 \cdot 24}$$

- C_{oc} : Sensible load from occupancy .
- C_{il} : Load from lighting .
- C_{eq} : Load from equipment .
- The result is averaged over 168 hours (7 days $\times$ 24 hours) .

3. Key Parameters and Limits

- **Export Factor (k_{exp})**: For standard verification, $k_{exp} = 0$, meaning exported energy is not deducted from consumption .

- **Consumption Limits ($C_{ep,lim}$):**
 - **Residential ($C_{ep,nren}$):** Ranges from **20** (Zone α) to **43** (Zone E) $kWh/m^2 \cdot year$ for new buildings .
 - **Residential ($C_{ep,tot}$):** Ranges from **40** (Zone α) to **86** (Zone E) $kWh/m^2 \cdot year$ for new buildings .
 - **Tertiary:** Limits are calculated as a base value plus a factor of CFI (e.g., Zone D: $20 + 8 \cdot CFI$) .

HE1: Conditions for Energy Demand Control

This section ensures the quality of the thermal envelope to minimize energy demand for heating and cooling .

[Image of building thermal envelope diagram]

1. Global Heat Transmission Coefficient (K)

The weighted average of thermal transmittance of the envelope, including thermal bridges :

$$K = \frac{\sum H_x}{A_{int}} = \frac{\sum b_{tr,x} [\sum A_{x,i} U_{x,i} + \sum l_{x,k} \psi_{x,k} + \sum \chi_{x,j}]}{\sum b_{tr,x} A_{x,i}}$$

- H_x : Heat transfer coefficient of element x .
- A_{int} : Total thermal exchange area .
- $b_{tr,x}$: Adjustment factor (1 for exterior, 0 for adjacent buildings) .
- $U_{x,i}$: Thermal transmittance of the element .
- $\psi_{x,k}$: Linear thermal transmittance of thermal bridges .

2. Air Permeability (n_{50})

This formula estimates the air change rate at 50 Pa pressure (mandatory for new residential buildings $> 120m^2$) :

$$n_{50} = 0.629 \cdot \frac{(C_o \cdot A_o + C_h \cdot A_h)}{V_{int}}$$

- C_o : Flow coefficient for opaque elements (**16** for new, **29** for existing) .
- C_h : Flow coefficient for openings (from test data) .
- A_o, A_h : Areas of opaque and opening elements .

- V_{int} : Internal air volume .

3. Thermal Resistance of Adjacent Spaces (R_u)

For modeling unconditioned spaces (like attics) as a homogeneous layer :

$$R_u = \frac{A_i}{\sum(A_{e;k} \cdot U_{e;k}) + 0.33 \cdot n \cdot V_k}$$

- A_i : Surface area between interior and unconditioned space .
- $A_{e;k}$: Surface area between unconditioned space and exterior .
- $U_{e;k}$: Transmittance of the external element .
- n : Ventilation rate of the unconditioned space .

4. Solar Control ($q_{sol;jul}$)

Limits the solar gains in July to prevent overheating:

- **Residential:** Limit $\leq 2.00 \text{ kWh/m}^2 \cdot \text{month}$.
- **Tertiary:** Limit $\leq 4.00 \text{ kWh/m}^2 \cdot \text{month}$.

HE2: Conditions for Thermal Installations

This section mandates compliance with the **Regulation of Thermal Installations in Buildings (RITE)** to ensure efficiency, comfort, and safety . While specific formulas are contained within the RITE, the guide emphasizes:

- Use of efficient generation equipment .
- Insulation of distribution networks .
- System regulation and control (e.g., temperature ranges of 21-25°C) .

HE3: Conditions for Lighting Installations

Regulates the energy efficiency of artificial lighting installations while ensuring visual comfort .

1. Energy Efficiency Value of the Installation (VEEI)

$$VEEI = \frac{100 \cdot P}{S \cdot E_m}$$

- P : Total installed power (W) including lamps and auxiliary equipment .
- S : Lit surface area (m^2) .
- E_m : Maintained average illuminance (lux) .

2. Parameters and Limits

- **Maximum Power (P_{max})**: Limits range from **5 W/m^2** (parking) to **25 W/m^2** (for $E > 600$ lux) .
- **$VEEI_{lim}$** : Specific limits by usage, e.g., **3.0** for administrative/offices, **4.0** for classrooms/hospitals, **10.0** for hotel rooms .

HE4: Minimum Renewable Energy for DHW

Requires a percentage of Domestic Hot Water (DHW) demand to be covered by renewable sources .

1. DHW Energy Demand (Approximation)

$$D_{ACS} = V_{ACS} \cdot C_{H2O} \cdot \rho_{H2O} \cdot (60^\circ - T_{\text{water network}})$$

2. Heat Pump Performance (SCOP)

For a heat pump to be considered renewable for HE4 compliance:

- $SCOP_{dhw} \geq 2.5$ (Electrically driven heat pumps) .
- $SCOP_{dhw} \geq 1.15$ (Thermally driven heat pumps) .

3. Contribution Parameters

- **Contribution Requirement:**
 - **60%** renewable contribution if demand $< 5000 \text{ l/d}$.
 - **70%** renewable contribution if demand $> 5000 \text{ l/d}$.
- **Residual Energy Limit:** A maximum of **20%** of extracted energy can be counted as contribution for residential use .

HE5: Minimum Electrical Energy Generation

Mandates renewable electricity generation (typically Photovoltaic) for buildings with a constructed area $> 1000m^2$.

[Image of photovoltaic solar panel system]

1. Minimum Power to Install (P_{min})

The installed power must be the **lesser** of P_1 and P_2 :

$$P_1 = F_{pr,el} \cdot S$$

$$P_2 = 0.1 \cdot (0.5 \cdot SC - SOC)$$

2. Calculation Parameters

- S : Constructed surface area of the building .
- SC : Non-trafficable roof surface area .
- SOC : Roof area occupied by solar thermal collectors .
- $F_{pr,el}$ (**Production Factor**):
 - **0.005** for private residential use .
 - **0.01** for other uses .

HE6: EV Charging Infrastructure

Establishes minimum endowments for electric vehicle charging stations in building parking lots .

1. Required Endowments

- **Pre-installation (Cable Conduits):**
 - **100%** of spaces for private residential buildings .
 - **20%** of spaces for other uses .
- **Charging Stations (Private Non-Residential):**
 - **1 station per 40 spaces** (general rule) .
 - **1 station per 20 spaces** (public administration buildings) .
 - **Accessible Stations:** 1 station per 50 accessible spaces (or fraction) .